Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Doctor of Philosophy (PhD) in Computer Science
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PhD Research Proposal

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Mode / Duration	Full-time
Date	1 October 2026

PhD Application – Declaration and Approval

Declaration and Approval

Student Declaration

This is my research proposal. I declare that the work presented in this proposal is my own and that it has been prepared under the supervision of the supervisors named below.

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Table of Contents

List of Figures ii
Abstract 1
1. Introduction 2
1.1 Background of the Research 2
1.1.1 The brain-tumour pathway and the state of computational support 2
1.1.2 The Sri Lankan context and the case for a national registry 3
1.2 Problem Statement and Research Gap 4
1.2.1 Why accuracy alone is not enough 4
1.2.2 Identified research gaps 4
1.2.3 Problem statement 5
1.3 Aim, Objectives and Research Questions 5
1.3.1 Purpose and aim 5
1.3.2 Research Objectives 5
1.3.3 Research Questions 6
1.4 Scope, Limits of Autonomy and Limitations 6
1.4.1 Scope of the study 6
1.4.2 Limits of AI autonomy 7
1.4.3 Limitations and delimitations 7
1.5 Justification and Significance of the Research 8
2. Literature Review 9
2.1 Approach to the Review 9
2.2 Brain Tumours: Burden, Classification and the Care Pathway 9
2.3 Deep Learning for Brain-Tumour Analysis from MRI 9
2.4 Data-Efficient, Self-Supervised and Domain-Adapted Learning 10
2.5 Uncertainty Quantification and Calibration 10
2.6 Selective Prediction, Learning to Defer and Human-in-the-Loop AI 10
2.7 Agentic and Multi-Agent AI in Healthcare 11
2.8 Computational Pathology and Multi-Source Diagnosis 11
2.9 Federated and Privacy-Preserving Learning 11
2.10 AI-Assisted Surgical Planning 12
2.11 Evaluation, Ethics, Governance and Regulation 12
2.12 Synthesis: What Is Established and What Remains Open 12
3. Research Design and Methods 14
3.1 Research Design 14
3.2 The Agents: What They Are and What Each One Does 14
3.2.1 What “agent” means in this framework 14
3.2.2 The orchestrator 15

3.2.3 The six specialist agents..... 15
3.2.4 The clinician-validator..... 16
3.2.5 How the agents work together, step by step..... 17
3.3 How the Escalation Decision Is Made..... 19
3.4 Work Packages..... 20
3.5 Population and Study Sample..... 20
3.6 Hypotheses..... 21
3.7 Operationalisation and Measures..... 21
3.8 Sources and Collection of Data..... 22
3.9 Data Analysis..... 22
3.10 Ethical, Privacy and Governance Considerations..... 23
3.11 Technology Stack..... 24
4. Research Plan and Timeline (36 Months)..... 25
5. Expected Outcomes and Publications..... 26
6. Evaluation and Success Criteria..... 26
References..... 27
Declaration and Approval..... 34

List of Figures

Figure 1.1: The brain-tumour care pathway, the specialised AI agent that supports each stage, and the clinician who retains final authority throughout..... 3
Figure 3.1: System architecture. Specialised agents are coordinated by an orchestrator over a shared registry data layer; every route to a clinical action passes the clinician-validator..... 17
Figure 3.2: The agent workflow from input to final output. Each numbered step names one agent, what it receives, what it produces, and what it hands to the next agent. The orchestrator applies the escalation rule after every step, and escalation to a clinician is available at every step. 18
Figure 3.3: How the escalation decision is computed. A raw score is calibrated, converted into an uncertainty, and compared against a threshold derived from the clinical cost of error rather than chosen by hand..... 19
Figure 3.4: Federated training across participating hospitals. Only model parameters are exchanged; raw patient data never leaves an institution..... 23

Abstract

Brain tumours are among the most demanding cancers to manage. Diagnosis depends on expert reading of magnetic resonance imaging (MRI) and of tissue taken at biopsy and examined under a microscope; treatment is time-critical; surgery must remove the tumour while sparing the parts of the brain that control speech, movement and vision; and follow-up runs for years. In Sri Lanka this work is carried by a small number of specialist units, and, unlike breast, lung or oral cancer, there is no national brain-tumour registry linking scans, laboratory results, clinical history, treatment and outcomes in one maintained resource.

The primary scientific contribution is an uncertainty-aware, human-in-the-loop multi-agent artificial-intelligence (AI) framework for the clinical decision workflow. Instead of one large model, the system is a team of specialised software agents coordinated by an orchestrator. The central research idea is that every agent must measure how confident it is, in a form that can be trusted numerically, and on that basis decide whether to act on its own or escalate to a clinician, who always remains the final validator. This turns the familiar principle that AI should carry routine load while doctors keep authority into a mechanism that can be specified, tuned and measured.

The framework is built and validated inside Sri Lanka's first National Brain Tumour Patient Registry. Its agents span the pathway: data-efficient MRI analysis for small and inconsistent local datasets; an agent that reads digitised microscope slides of biopsy tissue; an agent that combines imaging and tissue evidence to support tumour type, severity grade and treatment recommendation; and a pre-operative surgical-planning agent, built as a proof of concept, that proposes a candidate low-damage approach with an explicit map of structures at risk. A privacy-preserving federated-learning protocol trains these models across hospitals without moving raw patient data. Throughout, the AI plans, predicts and flags; clinicians decide and act, and the surgeon always performs and validates the operation.

1. Introduction

1.1 Background of the Research

Artificial intelligence (AI) now supports diagnosis, treatment planning and clinical documentation at a scale no human reader could match [26,82], but the same capability makes unsupervised deployment hazardous: opacity, dataset bias, unclear accountability and the difficulty of detecting silent failure have all constrained adoption [56,58]. Two findings dominate. Performance degrades under dataset shift: a pneumonia-detection model generalised poorly outside its training hospitals because it had latched onto site-specific artefacts rather than disease [54,55]. And a model that reports a confident answer gives the clinician no signal that it has left the region in which it is competent [28,31].

This has made human-in-the-loop (HITL) AI the dominant design philosophy for safety-critical medical AI. HITL interleaves human judgement and machine computation rather than substituting one for the other, inserting control points at consequential moments so a clinician can validate, refine or override a recommendation using findings, comorbidity and patient preference no model can see. HITL configurations outperform both fully automated AI and clinician-only workflows on accuracy, safety, efficiency and satisfaction, because the two participants fail in uncorrelated ways [25]. Crucially, HITL is usually asserted as a principle rather than implemented as a mechanism; a rare exception coupled retinal analysis to an explicit referral decision [27]. The open engineering question, and the one this research addresses, is not whether a human should be in the loop, but when, on which cases, and on what calibrated evidence.

1.1.1 The brain-tumour pathway and the state of computational support

The pathway is long, information-dense and unforgiving. It runs from non-specific neurological symptoms and a scan, through multi-sequence MRI, biopsy and examination of the tissue under a microscope, laboratory tests on that tissue, a multidisciplinary decision, surgery and follow-on therapy, to years of surveillance scanning. Since 2021 the World Health Organization classification has been integrated, so the diagnosis combines tissue appearance with laboratory results [36], making the task inherently multi-source. Brain tumours carry disproportionate mortality and affect working-age adults and children [37]. Surgery must balance removing as much tumour as possible against preserving the regions controlling speech, movement and vision: how much is removed is repeatedly associated with survival [38,39], while removing too much healthy tissue causes permanent loss of function. The pathway is also dependent: an error in outlining propagates into grading, grading into treatment selection, and treatment selection into outcome.

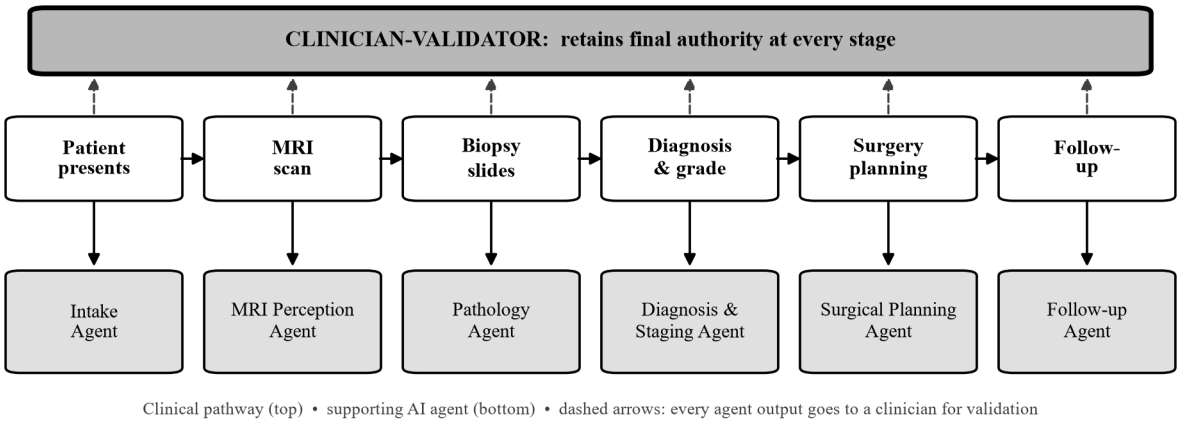


Figure 1.1: The brain-tumour care pathway, the specialised AI agent that supports each stage, and the clinician who retains final authority throughout.

Computational support has advanced rapidly at the level of individual sub-tasks. Deep learning dominates detection, outlining and classification on MRI, benchmarked on BraTS [40,41,42], with reviews documenting substantial gains [1,2,11], nnU-Net establishing strong baselines [43] and good outlining attainable with fewer sequences [12]. Computational pathology reaches clinical-grade analysis of whole digitised slides [44,45,46], extended to brain-tumour classification and grading [18,19,20,21]; combining tissue appearance with laboratory markers outperforms either alone [47], and near-real-time diagnosis during surgery has been demonstrated [48]. Yet individual models solve individual tasks: no accepted architecture assembles them into a supervised workflow with defined accountability. Two further shifts frame this work: agentic AI, whose safety, evaluation, traceability and oversight remain unresolved [3,4,5,60], and federated learning [49], which lets institutions train a shared model without disclosing raw data [50,51,52], proven across 71 sites on six continents [53] and where site data differ markedly [6,7,8,13].

1.1.2 The Sri Lankan context and the case for a national registry

GLOBOCAN 2024 estimates approximately 700 new brain and nervous-system cancer cases and around 587 deaths annually in Sri Lanka, with a five-year prevalence of roughly 2,500 [9]. Deaths are high relative to new cases and care is concentrated in a few tertiary units served by a limited pool of specialists. A National Cancer Registry exists for incidence and mortality reporting [10], but there is no imaging- and tissue-linked brain-tumour registry; patients are tracked in fragmented, institution-specific records. Clinically this impedes continuity of care, multidisciplinary review and outcome auditing. Scientifically it makes locally valid AI almost impossible to build, because models trained on foreign cohorts do not transfer reliably to Sri Lankan scanners, protocols, laboratory practice or case mix [2,54,55]. The project therefore separates a platform layer, the registry itself, which is the real-world deliverable giving the research authentic data and users; and a research layer, the focus of this PhD, which is the design, formalisation and validation of the AI framework that makes such a platform safe, data-efficient and privacy-preserving.

1.2 Problem Statement and Research Gap

1.2.1 Why accuracy alone is not enough

A model that is 95% accurate is not automatically safe: the danger lies in the 5% of confident mistakes, and in the absence of a mechanism by which the system can recognise the limits of its own competence. Modern deep networks are systematically over-confident, and their raw probabilities are poor estimates of correctness unless explicitly calibrated [31], a deficiency that worsens precisely when a model moves to a new hospital [71]. The issue is not a shortage of accurate models but the absence of a governing mechanism deciding, case by case, whether an output is fit to be acted upon. Where specialist review time is the scarcest resource, that mechanism also determines whether AI reduces clinician workload or merely relocates it.

1.2.2 Identified research gaps

No principled policy for when to defer. Clinical AI is deployed either as a fully manual aid, in which every output is reviewed, or as unconditional automation, in which none is. The primitives exist (selective classification [33], learning to defer [34,35], calibration [31], Bayesian and ensemble uncertainty [29,30,70] and conformal guarantees [32,75]), but were built for single models making single decisions, not assembled into a policy accounting for asymmetric error cost, finite expert review time [76], or a chain of dependent sub-decisions [4,28].

Unqualified hand-offs between agents. Agentic systems chain tasks, but auditable hand-offs, principled propagation of uncertainty and guaranteed points of human validation are unsolved for safety-critical medicine [3,5,60]. Errors then compound invisibly: a badly outlined lesion silently becomes a confident but wrong grade, then a wrong recommendation, and the composite decision is attributable to no participant, which is incompatible with clinical governance and with regulatory expectations of meaningful oversight [59].

Fusion that does not survive real data. Diagnosis, grade and treatment depend on integrating imaging and tissue analysis with laboratory markers where available [36], yet the two are handled in isolation. Fusion outperforms single-source models in research settings [47], but robust fusion where one source is missing, delayed or poor remains unsolved [18,19] and largely unexplored for low-resource settings.

Methods that assume abundant, standardised data. State-of-the-art models presuppose large, curated, protocol-standardised datasets [41,42]. Sri Lankan units hold small, heterogeneous collections, so out-of-the-box models under-perform and performance is fragile and scanner-dependent [2,11,54]. Label-efficient formulations exist [45,46,64,68,69] but are unvalidated on South Asian cohorts.

A chicken-and-egg problem in data sharing. Confidentiality, governance and absent data-sharing agreements make centralisation difficult, which starves training. Federated learning offers a route [49,51,52,87] proven at international scale [53], but the regime that matters here, a handful of hospitals holding small cohorts that differ markedly, is exactly where federated aggregation is least well characterised [6,7,8,88,89].

No way to measure the quality of the escalation decision. Clinical AI is reported chiefly through discriminative metrics on retrospective test sets [97]. Standards for early-stage clinical evaluation now require measurement of human factors and safety in situ [57,95,96], and ethical guidance

requires autonomy and accountability to be demonstrable rather than declared [58,59]. No established protocol quantifies whether a system defers on the cases where deferral was warranted.

1.2.3 Problem statement

These gaps identify the central research problem as the absence of a clinically responsible, formally governed multi-agent AI framework that can support the brain-tumour pathway across imaging, tissue analysis, combined diagnosis and pre-operative surgical planning while explicitly managing uncertainty and preserving clinician authority. Such a framework must determine, for each case and each sub-decision, when an agent can provide reliable decision-support information and when the case should be escalated to a qualified clinician. It must remain robust to the small, heterogeneous, non-standardised datasets characteristic of Sri Lankan institutions, and enable privacy-preserving collaboration without centralising raw patient data.

1.3 Aim, Objectives and Research Questions

1.3.1 Purpose and aim

The purpose of this study is to develop and evaluate a clinically responsible human-in-the-loop AI framework in which human oversight is implemented as an explicit, measurable, uncertainty-aware mechanism rather than treated only as a guiding principle, in a resource-constrained setting where datasets are limited, heterogeneous and distributed across institutions. The aim is to design, develop and evaluate an uncertainty-aware, human-in-the-loop multi-agent AI framework for brain-tumour care, integrating imaging, tissue analysis, combined clinical decision support and surgical planning, and to evaluate its feasibility within a national brain-tumour registry framework in Sri Lanka.

1.3.2 Research Objectives

The objectives are deliberately limited to four. The scope is kept controllable at this stage and can be broadened later in response to examiner and reviewer comment.

O1. To design and develop a multi-agent AI architecture for the brain-tumour care pathway (spanning patient intake, MRI analysis, tissue analysis, diagnosis and staging, surgical planning and follow-up), in which an orchestrator routes tasks, propagates uncertainty across agent hand-offs, and maintains an auditable record of every AI output and every clinician validation.

O2. To develop and evaluate a calibrated, uncertainty-aware escalation mechanism that determines when an agent may provide a decision-support output and when a case or sub-decision must be referred to a clinician for validation, based on selective prediction and learning-to-defer principles and on the clinical cost of error.

O3. To develop and evaluate the data-efficient models the agents depend on (for brain-tumour detection, outlining, classification and patient-specific anatomical reconstruction from MRI; for analysis of digitised biopsy slides; and for their combination with available laboratory markers) on small, heterogeneous Sri Lankan data, including training them across institutions by privacy-preserving federated learning without centralising raw patient data, and to benchmark their performance and calibration against established approaches.

O4. To clinically evaluate the proposed human-in-the-loop framework, covering its diagnostic and decision-support performance, the appropriateness of its escalation decisions, clinician

workload, usability, trust and safety, with surgical planning evaluated as a proof-of-concept decision-support component.

1.3.3 Research Questions

The research questions correspond one-to-one with the objectives and are likewise held to four, so that each can be answered to doctoral depth within the study period.

RQ1. How can specialised AI agents be orchestrated across the brain-tumour care pathway so that uncertainty is propagated between them, information is coordinated, and an auditable record of AI outputs and clinician validations is maintained?

RQ2. How can uncertainty estimation and calibration be incorporated into that framework to determine when the system should provide a decision-support output and when it should defer to a clinician?

RQ3. How effectively can data-efficient and multi-source methods detect, outline, classify and grade brain tumours from small and heterogeneous Sri Lankan MRI and biopsy-slide data, including under federated training across hospitals, and how do their performance and calibration compare with established approaches?

RQ4. To what extent can the proposed human-in-the-loop framework improve clinically relevant decision support while maintaining appropriate clinician involvement, acceptable workload, usability, trust and safety, including the feasibility of MRI-based surgical-planning support?

1.4 Scope, Limits of Autonomy and Limitations

1.4.1 Scope of the study

Depth of investigation is deliberately differentiated across three tiers. The first and primary tier is the uncertainty-aware orchestration and escalation mechanism, supported by data-efficient MRI models; this is the central methodological contribution. The second is computational pathology and combined decision support, remaining within the role of decision support rather than autonomous diagnosis. The third is privacy-preserving federated learning. The surgical-planning component is a proof-of-concept module covering patient-specific anatomical reconstruction, outlining of the tumour and of structures at risk, and visualisation of surgical risk; it provides no autonomous surgical decisions. The registry serves as enabling infrastructure for registration, identification, longitudinal monitoring, follow-up and secure management of clinical information. The study is limited to participating Sri Lankan institutions, and clinically to tumours included in the registry under the predefined eligibility criteria.

1.4.2 Limits of AI autonomy

Autonomy is conditional, uncertainty-aware and task-specific rather than absolute. The system is not designed to independently diagnose, determine treatment or perform surgical procedures. Each agent generates a decision-support output with an uncertainty estimate; cases exceeding predefined thresholds are escalated, and the final clinical decision remains the responsibility of the treating professional. On surgery the position is explicit and non-negotiable: the system never performs or controls an operation. It plans, predicts, flags and recommends; the neurosurgeon executes and validates every consequential step. The architecture enforces clinician validation at every decision point rather than relying on procedural convention [58,59]. Three boundaries follow. Bounded decision authority: the AI issues no diagnosis, grade or treatment decision of record, only evidence, a calibrated confidence and a recommendation a clinician accepts, amends or rejects. Conservative default: under uncertainty or novelty the system defers rather than acts. Transparency of basis: every autonomous action carries the evidence and confidence behind it, plus a model-facts style statement of the conditions under which the component was validated [100], so a reviewer can reconstruct not only what the system did but why it did not ask [56].

1.4.3 Limitations and delimitations

Component performance may vary across tumour types, imaging protocols, institutions and patient populations, and the federated component depends on the number and characteristics of participating institutions, computing infrastructure and connectivity. Several adjacent areas lie outside scope and are recorded as future directions: guidance during surgery, robotic actuation and real-time navigation; development of new laboratory assays, existing marker results being consumed as input rather than generated; physics-based simulation of tissue deformation, radiotherapy dose planning and chemotherapy optimisation; paediatric-specific modelling as a separate strand, though paediatric cases are not excluded from the registry; and deployment beyond the participating units and certification as a medical device, though the evidence generated is intended to support a later certification pathway.

Three further limitations should be stated plainly. Cohort size: national incidence means the registry will accrue hundreds rather than thousands of fully linked cases, bounding subgroup analyses and making data-efficient methods a necessity. Reference-standard dependence: clinician-confirmed diagnoses serve as ground truth, so the framework is evaluated against expert consensus rather than an independent biological gold standard. And the study is not a randomised trial of patient outcomes; validation measures diagnostic performance, workload, escalation appropriateness, safety and trust in an assistive workflow [57], so claims about survival benefit are explicitly not made. The work assumes ethics approval and data-sharing agreements can be secured before data access, that units can supply usable MRI and digitisable biopsy material, that clinician time is available albeit scarce, and that units will operate shared governance for federated training; each assumption is paired with a mitigation in the risk register.

1.5 Justification and Significance of the Research

Scientific and clinical. How an AI agent should decide when it is competent to act and when it must defer is a computer-science problem before it is a clinical one, sitting at the intersection of uncertainty quantification [29,30,32,70], calibration [31,71], selective prediction and learning to defer [33,34,35,77] and multi-agent orchestration [3,5,79]. The literature treats these as separate concerns and treats oversight as a principle asserted in prose rather than a mechanism specified, calibrated and measured. Formalising selective autonomy over a chain of dependent sub-decisions, and evaluating it on an explicit workload-versus-safety frontier, is methodologically novel and transferable well beyond this clinical domain. Brain-tumour care is an unusually strong test case: it requires formal integration of several evidence sources [36]; uncertainty must propagate along a sequential chain; the cost of error is severe and asymmetric [38,39]; and specialist review time is the binding constraint, so reduced review load is a directly meaningful outcome.

National and methodological. Sri Lanka has no imaging- and tissue-linked brain-tumour registry, and its absence limits clinical continuity, outcome auditing and research capacity simultaneously [9,10]. Establishing the first such registry delivers durable value to the participating units independently of the research findings, and the associated open, de-identified benchmark addresses the near-total absence of South Asian representation in public brain-tumour datasets [41,42]. The constraints of the setting are also what make the research generalisable: small, heterogeneous data and barriers to centralisation are the normal condition of clinical AI in most of the world, so methods demonstrated under scarcity, heterogeneity and federation reach further than those requiring large curated corpora [50,51,53]. Evaluating on local data guards against the site-specific shortcut learning that has undermined models validated only internally [54,55].

Ethical, regulatory and feasible. International guidance places human autonomy, transparency and accountability among its core principles for AI in health [58], and regulation now requires that high-risk systems be designed so people can oversee them and intervene [59]; explainability alone is insufficient without an account of responsibility [56], and algorithmic bias arises from design choices never examined [98]. A framework making oversight measurable and auditable rather than merely declared addresses a requirement that is becoming mandatory. The enabling components are mature enough to build on but immature enough to contribute to [29,30,32,44,45,84,85,3,5,60,53,6,7,8]. The project combines established computer-science supervision with clinical supervision, giving direct access to the patient pathway, to authentic data and to the clinicians who will validate and use the system; the tiered scope, front-loaded ethics approval and development on public benchmarks while local data accrue make the plan deliverable within 36 months.

2. Literature Review

2.1 Approach to the Review

This chapter reviews the literature the framework builds on and identifies, in each area, the point at which existing work stops. It follows the architecture of the proposed system: the clinical domain, the perception layer, the core contribution of uncertainty and deferral, orchestration, computational pathology, federated learning, surgical planning, and evaluation and governance. Sources were drawn from the major venues for medical image analysis, machine learning and clinical informatics, with recent systematic and scoping reviews [1,2,3,5,11]. Priority was given to work from 2015 onward, to studies reporting external validation, and to sources reporting negative results, since these are more informative about deployment risk.

2.2 Brain Tumours: Burden, Classification and the Care Pathway

Brain and nervous-system tumours are uncommon but carry disproportionate mortality, with poor outcomes concentrated in the most aggressive types [37]. Sri Lanka sees roughly 700 new cases and 587 deaths annually, with a five-year prevalence of about 2,500 [9,10]. The most consequential recent change for computational work is classificatory: the 2021 World Health Organization classification integrates tissue appearance with laboratory marker results to define tumour type and grade [36]. A model trained on a pre-2021 label is therefore not predicting the same target as one aligned to the current classification, and a definitive diagnosis is now multi-source by definition, so single-source systems can only support a diagnosis, never produce one of record. The pathway is a dependent chain in which errors propagate forward and the cost of error is strongly asymmetric [38,39].

2.3 Deep Learning for Brain-Tumour Analysis from MRI

U-Net [61] and its volumetric extension [62] remain the structural basis of most published systems, and shared benchmarks drove the field: BraTS established a common multi-institutional dataset, annotation protocol and evaluation methodology [40,41,42]. Transformer designs such as Swin UNETR followed [63], and nnU-Net showed that careful automated configuration of a conventional architecture matches or exceeds many bespoke designs [43]. Reviews document rapid gains [1,2,11], and strong outlining remains achievable with fewer MRI sequences [12]. The limitation is generalisation: performance depends on large, curated, protocol-standardised data and stays fragile on small single-institution datasets and unseen scanners, where models can learn site-specific artefacts rather than disease [2,11,54,55]. For a Sri Lankan deployment this is the principal technical risk to the perception layer, and it motivates RQ3.

2.4 Data-Efficient, Self-Supervised and Domain-Adapted Learning

Because annotated local data will be scarce, the perception layer depends on methods that reduce label dependence. Contrastive learning [64,65], self-distillation such as DINO [66] and masked autoencoding [67] showed that useful representations can be learned from unlabelled images alone, and these transfer to medicine with measurable benefit, improving classification when labels are few [69] and outlining when pre-trained on unlabelled 3D volumes [68]. The benefit is largest exactly where labels are scarcest, and pre-training can use unlabelled local scans, which carry the scanner characteristics transferred models lack. What remains unestablished is whether the resulting representations yield confidence estimates fit to drive an autonomy decision; most evaluations report only accuracy on curated benchmarks. That intersection is where RQ3 meets RQ2.

2.5 Uncertainty Quantification and Calibration

The premise of the framework is that an agent must know when it does not know, and the foundations are established. Monte-Carlo dropout approximates Bayesian inference in deep networks [29]; deep ensembles are a simple, strong alternative [30]; and separating uncertainty caused by noise from uncertainty caused by limited knowledge clarifies which component more data can reduce [70]. Modern networks are systematically over-confident, and post-hoc calibration such as temperature scaling substantially improves the match between predicted probability and observed correctness [31], while conformal prediction gives distribution-free coverage guarantees [32,75]. Medical imaging has adopted these with mixed results [72,73,74]. The most important negative result is that calibration degrades precisely under dataset shift, exactly where a reliable confidence signal is most needed [71]. The gap is not the absence of estimators but their disconnection from action [28].

2.6 Selective Prediction, Learning to Defer and Human-in-the-Loop AI

Selective classification equips a model with a reject option and characterises the risk-coverage trade-off [33]. Learning to defer generalises this by modelling the downstream human: the system learns a joint policy over predictor and rejector that accounts for the expert's own error profile [34,35]. Related work frames the problem as allocating scarce human effort through triage [76] and as learning to complement rather than imitate human decision-makers [77]. Human-factors work qualifies these results: explanations can increase acceptance of incorrect recommendations [78]. In healthcare, human-in-the-loop configurations outperform both fully automated and clinician-only workflows [25], and control points, escalation and confidence-aware deferral are prerequisites for real use [4]; yet deployed exemplars remain rare [27]. Three limitations define the core gap: these formulations address a single model making a single decision, not a chain in which uncertainty must be composed across agents; they assume a fixed, homogeneous cost of error, whereas clinical costs are asymmetric and vary by decision type; and the expert is modelled as an always-available oracle, whereas specialist capacity is the binding constraint. Addressing these jointly is the substance of RQ1 and RQ2.

2.7 Agentic and Multi-Agent AI in Healthcare

Interest has shifted to agentic systems whose components reason over multiple steps, invoke tools and coordinate with one another, enabled by chain-of-thought prompting [80] and the interleaving of reasoning with tool use [79]. In medicine, large language models encode substantial clinical knowledge [81], the case for generalist medical AI has been argued [82], and conversational diagnostic systems have been evaluated against clinicians [83]. Surveys map the application space while consistently identifying the same unresolved issues: safety, evaluation methodology, traceability of multi-step reasoning, and accountability when a composite decision emerges from several components [3,4,5,60]. The gap is structural: existing work concentrates on capability and end-to-end task success, not on how uncertainty is represented at the interface between agents, how a downstream agent should behave when an upstream input is unreliable, or how an auditable record should be maintained. No published architecture makes uncertainty a first-class element of the contract between agents, which is the contribution proposed under RQ1.

2.8 Computational Pathology and Multi-Source Diagnosis

Clinical-grade performance was demonstrated using weakly supervised learning on digitised slides at scale [44], with attention-based multiple-instance learning letting a single slide-level label supervise attention to individual regions [45,46]. Pathology foundation models now serve as general-purpose encoders [84,85,86], transferring strongly to tasks with few local labels. In brain-tumour work, deep learning supports grading and classification [18,19,20,21], and application during surgery matched conventional interpretation prospectively [48]. Because the 2021 classification is integrated [36], the decisive question is fusion: combining tissue appearance with laboratory features improves diagnosis and prognosis relative to either [47]. What remains unsolved is robustness to real-world incompleteness: published fusion models assume every source present and of consistent quality, whereas in practice tissue analysis lags imaging, slide quality varies, and markers may be unavailable for cost reasons. Fusion that degrades gracefully is not established, and is part of RQ3.

2.9 Federated and Privacy-Preserving Learning

Federated learning enables training across institutions without centralising raw data, using iterative local training and parameter aggregation [49]. Its medical relevance was established early [50,51,52,87], and the decisive demonstration trained a glioblastoma tumour-boundary model across 71 sites on six continents, improving substantially on any single site's model without disclosure of raw patient data [53]. Aggregation across very different sites has been addressed through proximal regularisation [88] and variance reduction correcting for client drift [89], and privacy strengthened through differentially private training [90,91]; recent work adds encryption, reputation-aware aggregation and heterogeneity handling [6,7,8,13,14]. Two gaps persist. The regime of a few hospitals each holding a small, markedly different cohort is under-characterised, and is where aggregation instability and client drift are most severe. And federation is studied as a training-time concern in isolation: how federated updates interact with calibration, and therefore with an escalation policy validated on a previous model version, is essentially unexamined. A model that gains accuracy while losing calibration would silently change deployed safety behaviour, making this a first-order concern for RQ3.

2.10 AI-Assisted Surgical Planning

Pre-operative planning support has developed along two lines: patient-specific anatomical reconstruction from imaging, supported by mature open platforms [94] and digital-twin formulations [15,16,17]; and algorithmic trajectory planning, formulating the approach path as a constrained optimisation over anatomy to be avoided, founded on sampling-based optimal motion planning [92], with clinical instances computing electrode trajectories [93] and risk-aware planning seeking corridors that avoid critical tissue and blood vessels [22,23,24]. This literature establishes feasibility and its own boundaries: reconstruction quality depends on the outlining of structures at risk, itself an uncertain perception task, and evaluation is typically geometric rather than clinical acceptability judged by operating surgeons. Crucially, published systems present a single recommended trajectory without stating confidence in the underlying anatomical delineation, which is why the surgical-planning agent is scoped as a proof-of-concept demonstrator under RQ4.

2.11 Evaluation, Ethics, Governance and Regulation

Methodological guidance distinguishes internal from external validation and warns against inflated claims from inappropriate test-set construction [97]. Reporting standards followed: CONSORT-AI for randomised evaluations [95]; TRIPOD+AI for prediction-model reporting [96]; and DECIDE-AI for the stage most relevant here, small-scale clinical evaluation of AI decision support, which explicitly requires reporting of human factors, of the clinician-system interaction, and of safety in situ [57]. World Health Organization guidance places human autonomy, transparency and accountability among its core principles [58]; interpretability alone cannot discharge accountability without an account of responsibility [56]; algorithmic bias has been shown to arise from an apparently reasonable choice of prediction target [98]; model-facts labelling communicates validation conditions [100]; and maintenance dominates the lifetime cost of deployed systems [99]. Regulation has made this concrete: the European Union's Artificial Intelligence Act classifies medical AI as high-risk and requires, in Article 14, that such systems be designed so people can oversee them, remain aware of automation bias, and intervene or interrupt operation [59]. Yet none of these standards specifies how to measure the quality of an escalation decision itself, and producing such a protocol is part of the contribution claimed here.

2.12 Synthesis: What Is Established and What Remains Open

Four things are established: the individual perception tasks the framework requires are technically solved to a clinically useful standard on curated data, for MRI [1,2,40,42,43] and digitised slides [44,45,84]; the tools for quantifying and calibrating predictive uncertainty are mature [29,30,31,32], with formal mechanisms for acting on it [33,34,35]; federated learning is a proven route to multi-institutional training without data sharing [50,53]; and both ethical guidance and binding regulation now require demonstrable human oversight of high-risk clinical AI [58,59].

Four things remain open, and together they define this research. The primitives for selective autonomy have never been assembled into a calibrated escalation policy for a dependent chain of clinical sub-decisions with asymmetric error costs and finite expert capacity [33,34,76]. Agentic architectures do not treat uncertainty as part of the contract between agents, so hand-offs are unqualified and accountability for composite decisions is undefined [3,5,60]. Multi-source diagnosis is demonstrated on complete, high-quality cohorts rather than under the incompleteness of real practice [47]. And data-efficient, federated and calibrated learning have been studied

separately, leaving unexamined the interaction that matters most: federated updates and domain adaptation can change a model's calibration and therefore silently alter the safety behaviour of any escalation policy built on it [53,71,89]. The framework proposed here closes these four gaps in a single, clinically validated system.

3. Research Design and Methods

3.1 Research Design

This research produces an artefact (a working multi-agent clinical AI framework) and evaluates it against measurable criteria. The design is therefore design-science research, the established paradigm for computing research whose contribution is a constructed system evaluated in its intended environment [101], following the six-activity methodology of problem identification, objective definition, design and development, demonstration, evaluation and communication [102]. It is chosen over a purely experimental design because the central claim, that selective autonomy can be made calibrated and auditable, cannot be demonstrated by benchmarking alone. Methods are selected by fitness to each research question [103], yielding a mixed-methods design. The dominant quantitative strand comprises retrospective and simulated experiments on registry and public data, measuring outlining, classification, calibration, escalation and federation performance. The supporting qualitative strand comprises clinician feedback and interviews measuring trust, perceived usefulness and the acceptability of escalation behaviour. The two combine in an explanatory sequential arrangement in the final validation [103,104]. Work proceeds through iterative build-and-evaluate cycles closed by clinician feedback; because the artefact evolves, every result is tied to a versioned model and escalation policy, and none is carried forward across a change to either without re-evaluation.

3.2 The Agents: What They Are and What Each One Does

3.2.1 What “agent” means in this framework

In this proposal an agent is a self-contained software component with one clearly bounded job, a defined input, a defined output, and one additional obligation that distinguishes it from an ordinary module: it must report how confident it is in what it has just produced, on a scale calibrated so that the number can be believed. An agent is not a person, not a chatbot and not a general-purpose reasoner. It is closer to a specialist on a team who does one job, states how sure they are, and passes the result on.

Three consequences follow and shape the whole architecture. Because each agent does one job, it can be developed, tested and replaced independently of the others. Because each agent emits a confidence alongside its result, the component that coordinates them can reason about reliability rather than merely about outputs. And because confidence travels with the result across every hand-off, a later agent is never misled into treating a doubtful input as a certain one, which is the specific failure this framework is designed to prevent. The system comprises six specialist agents, one orchestrator that coordinates them, and the clinician, who is a participant in the workflow rather than an observer of it.

3.2.2 The orchestrator

The orchestrator is the coordinating component, and is not itself a diagnostic model. Its inputs are the case as it stands and the result and confidence just produced by whichever agent last ran; its outputs are a routing decision (which agent runs next, or whether the case stops here) and a record written to the audit trail. After every step it does four things: it reads the agent's calibrated confidence; it combines that confidence with the clinical cost of getting this particular kind of decision wrong; it applies the escalation rule of Section 3.3, producing one of three actions (proceed, gather more evidence, or escalate to a clinician); and it writes what happened, why, and with what confidence to an auditable record. The orchestrator is therefore where the scientific contribution of this thesis lives: the agents perform recognised tasks, but the policy governing when their outputs may be used without human review is the novel mechanism under study.

3.2.3 The six specialist agents

Each agent is described below by its role, what it receives, what it produces, and what it hands to the next stage. Every output includes a calibrated confidence, and every agent can be interrupted by the orchestrator and its case sent to a clinician.

1. Intake Agent. Role: to turn a messy real-world referral into one clean, standardised, de-identified patient record the rest of the system can rely on. Inputs: patient details, the referral report, the MRI study as it comes from the scanner, and the digitised biopsy slides where these exist. It identifies the patient, checks whether a record already exists so the same person is not registered twice, removes direct identifiers, and applies automated quality checks to the scans and slides. Outputs: a single standardised registry record, plus a quality flag on each item of data. Hands off to: the MRI Perception Agent; and, where a poor-quality or unreadable item is flagged, to the orchestrator, which requests a repeat or refers the case for human attention before any analysis begins.

2. MRI Perception Agent. Role: to find and outline the tumour on the scan and suggest what type it is. Inputs: the standardised multi-sequence MRI study from the Intake Agent. It detects whether a lesion is present, draws its boundary slice by slice, measures its volume, and proposes a tumour type. Because local scans are few and vary between machines, this agent is built with data-efficient methods and is where the calibration work of Objective 3 is concentrated. Outputs: the tumour location and outline, a suggested type, a volume measurement, and a calibrated confidence for each. Hands off to: the Pathology Agent and the Diagnosis and Staging Agent, which receive both the findings and the confidence attached to them; and to the Surgical Planning Agent, which later reuses the same outlines.

3. Pathology Agent. Role: to read the digitised microscope slides of the biopsy and produce supporting evidence about the tumour. Inputs: whole digitised slides of the tissue removed at biopsy. It analyses the slide at high magnification, identifies the regions carrying diagnostic information, and derives evidence for tumour type and severity grade, using methods that need only a label for the slide as a whole, because region-by-region expert annotation is not available at the scale required. Outputs: evidence for tumour type and grade, a map showing which regions of the slide drove that conclusion, and a calibrated confidence. Hands off to: the Diagnosis and Staging Agent. If the slide is of poor quality, or the evidence conflicts with the MRI findings, the conflict itself is a trigger for escalation.

4. Diagnosis and Staging Agent. Role: to bring the separate lines of evidence together into one supported conclusion. Inputs: the MRI findings and confidence from Agent 2, the tissue findings and confidence from Agent 3, and any laboratory marker results recorded for the patient. Because the current international classification defines the diagnosis by combining tissue appearance with laboratory results, this is the agent at which a diagnosis can first be supported at all. It must also cope with the routine case in which one source is missing, delayed or poor; it degrades gracefully rather than failing, and reports how much of its confidence came from each source. Outputs: a supported tumour type, a severity grade, a treatment-related recommendation, a breakdown of which evidence contributed what, and a combined calibrated confidence. Hands off to: the clinician for validation in every case where the decision is consequential, and, once validated, to the Surgical Planning Agent and the Follow-up Agent.

5. Surgical Planning and Risk-Path Agent. Role: to help the surgeon see, before an operation, what the anatomy looks like and which route carries least risk. Inputs: the MRI outlines from Agent 2 and the validated diagnosis from Agent 4. It reconstructs a three-dimensional model of that specific patient's anatomy, outlines the structures that must be avoided (the regions controlling speech, movement and vision, major blood vessels, and the nerve pathways between them) and computes a candidate approach route minimising expected damage. Crucially it also carries forward the uncertainty in the underlying outlines, so a route based on a doubtful boundary is presented as such rather than as a confident recommendation. Outputs: a patient-specific 3D model, a map of structures at risk, a candidate approach route, and an explicit statement of confidence in the anatomy the route was computed against. Hands off to: the neurosurgeon, always. This agent is a proof-of-concept demonstrator; it never performs, controls or guides an operation.

6. Follow-up Agent. Role: to keep track of the patient after treatment. Inputs: the validated record and any later surveillance scans. It schedules surveillance, compares each new scan against the previous one, and measures change in tumour size or appearance over time. Outputs: a surveillance schedule and a report of longitudinal change, with a confidence attached to each measured change. Hands off to: the clinician, where a change exceeds a threshold or the comparison is uncertain, and to the registry, which retains the record as part of the national longitudinal resource.

3.2.4 The clinician-validator

The clinician is treated as a component of the workflow with a defined role, not a reviewer appended to the end of it. Inputs: any case the orchestrator escalates, together with the agent's output, the evidence behind it, and the confidence that triggered the escalation. Outputs: an accept, amend or reject decision, which becomes the decision of record. Two invariants hold throughout: there is no route from any agent to a clinical action of record that does not pass a clinician, and every escalation, every autonomous action and every clinician decision is written to the audit trail.

3.2.5 How the agents work together, step by step

Figure 3.2 traces one case from arrival to final output. The case enters at the Intake Agent and moves down the chain; at each step the agent produces its result and its confidence, the orchestrator applies the escalation rule, and the result is either passed on, returned for more evidence, or sent to a clinician. Nothing crosses a hand-off without its confidence attached, and no step can reach a clinical action without passing the clinician-validator. Figure 3.1 shows the same system as a static architecture, with the agents coordinated by the orchestrator over the shared registry data layer that also holds the audit trail.

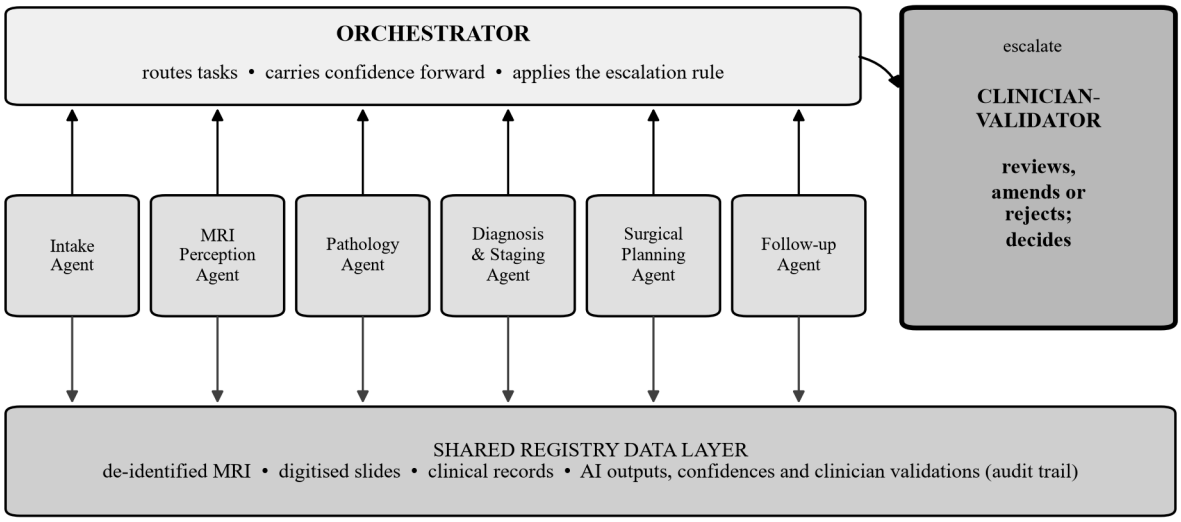


Figure 3.1: System architecture. Specialised agents are coordinated by an orchestrator over a shared registry data layer; every route to a clinical action passes the clinician-validator.

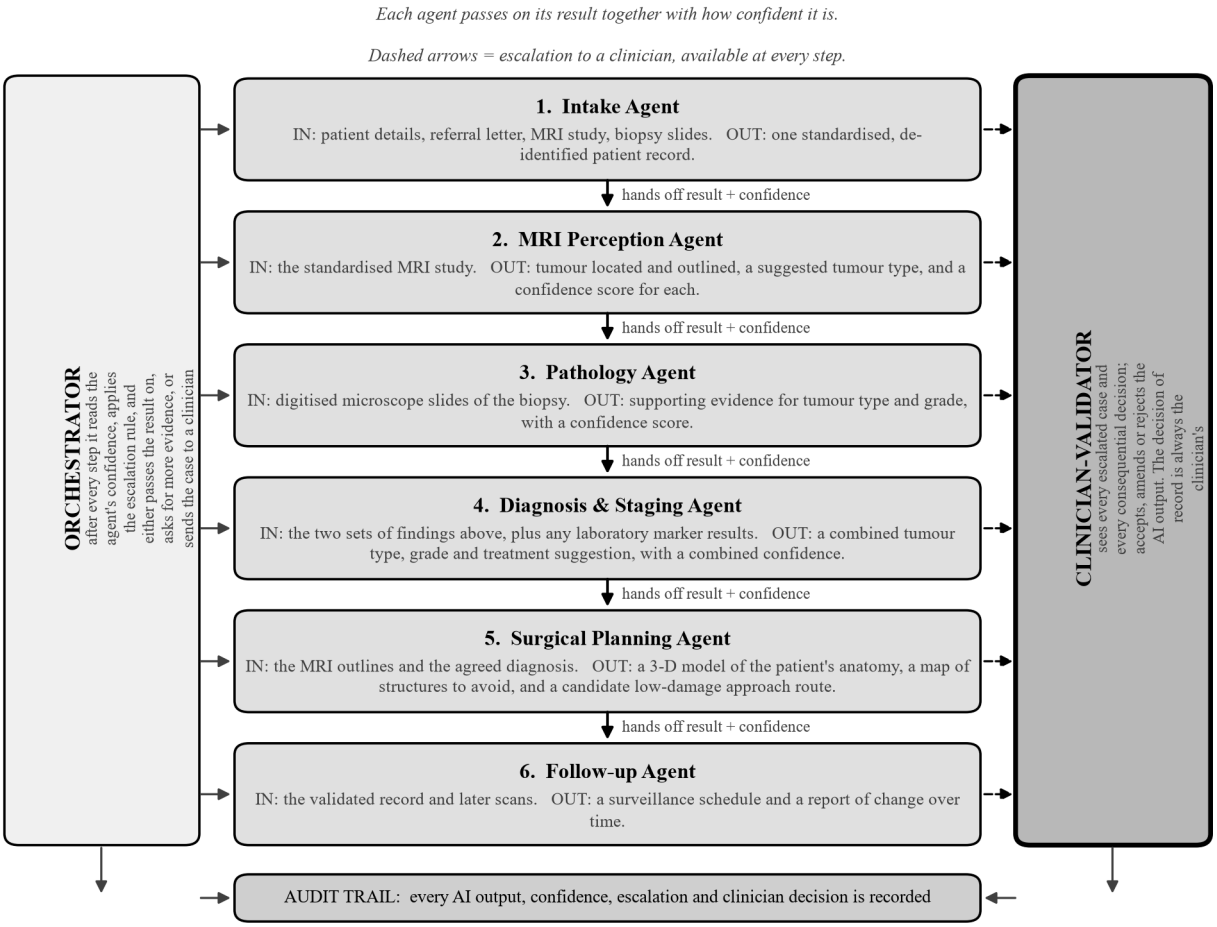


Figure 3.2: The agent workflow from input to final output. Each numbered step names one agent, what it receives, what it produces, and what it hands to the next agent. The orchestrator applies the escalation rule after every step, and escalation to a clinician is available at every step.

3.3 How the Escalation Decision Is Made

The escalation rule is the mechanism under study, and is deliberately simple to state. An agent's raw output score is first calibrated, so a reported probability of 0.8 corresponds to being right about 80% of the time rather than to an arbitrary number produced by the network. The calibrated probability is converted into an uncertainty, then compared not against a threshold chosen by hand but against one derived from the clinical cost of the two kinds of mistake available at that decision, because wrongly reassuring a patient and wrongly alarming one are not equally harmful, and the ratio between those costs is what should set the operating point. The comparison yields exactly one of three actions: proceed, gather more evidence, or escalate to a clinician. The costs are elicited from the clinical supervisor and participating clinicians rather than assumed, and a list of decisions that are always escalated regardless of confidence is fixed in advance.

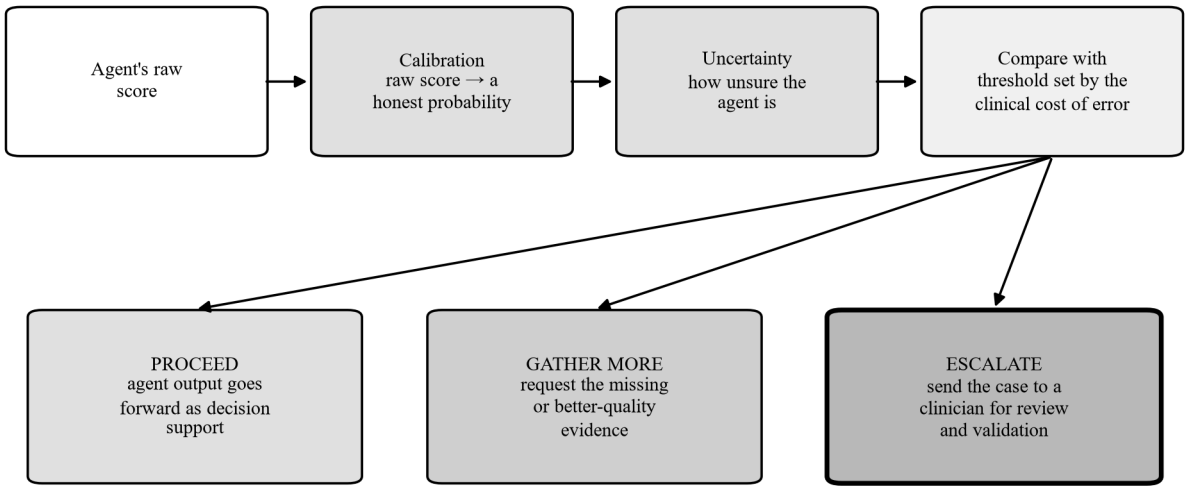


Figure 3.3: How the escalation decision is computed. A raw score is calibrated, converted into an uncertainty, and compared against a threshold derived from the clinical cost of error rather than chosen by hand.

As variables, the independent variables are the design choices under the researcher's control: uncertainty-estimation method, calibration method, escalation policy and its cost parameters, training regime (local, transfer, self-supervised, federated) and fusion strategy. The mediating variable, and the theoretical heart of the framework, is the calibrated uncertainty attached to each agent output, since the thesis claims that the effect of those choices on outcomes is transmitted through the quality of this estimate rather than through raw accuracy. The dependent variables are diagnostic performance, clinician workload, escalation appropriateness, safety and trust. Moderating variables recorded for every case are scanner and acquisition protocol, image and slide quality, tumour type and grade, completeness of evidence, and site of origin.

3.4 Work Packages

The work is organised into seven work packages aligned with the objectives. WP1 delivers the registry platform and data pipeline. WP2 delivers data-efficient MRI perception. WP3, the scientific core, delivers uncertainty-aware orchestration and the escalation policy (RQ1 and RQ2). WP4 delivers computational pathology and combined diagnosis and staging. WP5 delivers federated learning across hospitals. WP6 delivers the surgical-planning agent as a proof of concept. WP7 delivers clinical validation. Each follows the same internal cycle of design, implementation, quantitative evaluation and clinical review. WP2, WP4 and WP5 together answer RQ3; WP6 and WP7 together answer RQ4.

3.5 Population and Study Sample

The target population comprises all patients with primary or secondary brain and nervous-system tumours in Sri Lanka, estimated at approximately 700 new cases annually and a five-year prevalence of about 2,500 [9]. The accessible population comprises patients presenting to the participating tertiary units during the study period and meeting the eligibility criteria, recruited by consecutive sampling. The expected cohort is several hundred patients; the exact number depends on incidence, referral patterns and data availability, and the number with complete linked imaging, tissue, marker, treatment and follow-up information will be smaller. Patients are eligible if they are of any age with a suspected or confirmed brain or nervous-system tumour, managed at a participating institution during the study period, with at least one clinically relevant MRI examination in usable standard scan-file form, sufficient clinical information to establish the reference standard, and appropriate ethical approval and consent or waiver. Records are excluded where imaging is unusable and unrecoverable, where reference-standard information cannot be established, where de-identification cannot be completed, or where data-governance requirements are not met. Patients are not excluded solely for missing tissue, marker or follow-up information when they remain eligible for an individual component; they contribute according to data availability, which is what allows evaluation under the incomplete-data conditions expected in real practice.

A second population comprises the clinicians involved in diagnosis and management at participating institutions: consultant neurosurgeons, radiologists, histopathologists and experienced senior trainees. Because this group is small and highly specialised, clinicians are recruited by purposive sampling based on expertise and are used primarily to evaluate interpretability, trust, usability, workload, escalation decisions and clinical usefulness. For the reader evaluation a multi-reader, multi-case design is used so the available experts are used efficiently: each assesses a common set of cases under assisted and unassisted conditions, with counterbalanced ordering and a washout period. A third population comprises public datasets used for pre-training, benchmarking and external comparison, including BraTS. Public data are not representative of the Sri Lankan population and are therefore not the sole basis for any clinical performance claim; their purpose is to establish reproducible baselines before sufficient local data become available.

3.6 Hypotheses

The hypotheses are testable statements aligned with the objectives and research questions. The corresponding null hypotheses assume no statistically significant difference between the compared methods or conditions.

H1 (RQ2). A calibrated, uncertainty-aware escalation policy will achieve a more favourable workload-safety trade-off than conventional fixed-confidence thresholding, shown by lower decision error at an equivalent escalation rate, or lower escalation at an equivalent error rate.

H2 (RQ1). Propagation of calibrated uncertainty between interacting agents will reduce downstream decision errors compared with conventional hand-offs that do not account for the reliability of upstream outputs.

H3 (RQ3). Self-supervised pre-training on unlabelled local MRI data followed by supervised fine-tuning will improve the performance and calibration of brain-tumour MRI models compared with transfer learning from public benchmark datasets alone.

H4 (RQ3). Combining MRI and biopsy-slide evidence will achieve higher diagnostic and grading concordance with the clinician-established reference standard than the best-performing single-source approach.

H5 (RQ3). Federated training across participating hospitals will improve generalisation compared with single-site training, while maintaining performance within a predefined margin of centrally trained models and preserving calibration within a predefined tolerance, as measured by expected calibration error and Brier score.

H6 (RQ4). The AI-assisted human-in-the-loop workflow will reduce clinician review time and improve perceived usability and trust compared with an unassisted workflow, while maintaining diagnostic performance within a predefined non-inferiority margin.

3.7 Operationalisation and Measures

Outlining quality is measured by Dice similarity coefficient, 95th-percentile Hausdorff distance, sensitivity and specificity, per case with confidence intervals [126]; classification and diagnosis quality by area under the receiver-operating characteristic curve, balanced accuracy, per-class F1 and concordance with the clinician reference standard. Calibration, the mediating variable, is measured by expected and maximum calibration error [135], the Brier score [136], reliability diagrams [31], and empirical coverage against the nominal level [32,75]. Selective autonomy is measured by coverage, the proportion of cases handled autonomously; selective risk, the error rate among those cases; the area under the risk-coverage curve [33]; the appropriate-escalation rate, the proportion of escalated cases in which the clinician changed the AI output; and the missed-escalation rate, the proportion of autonomously handled cases that blinded expert review judged to require human input, the primary safety indicator. Workload is measured as review minutes per case from the audit log and by the NASA Task Load Index [132]; trust and usability by the System Usability Scale [133] and a Likert instrument; inter-clinician agreement by kappa [131] and Krippendorff's alpha [140]. Federated indicators comprise the federated-versus-centralised accuracy difference, communication cost, calibration change after aggregation, the differential-privacy budget [90,91], and an audit-verified indicator that no raw patient data left any institution.

3.8 Sources and Collection of Data

Primary data comprise diagnostic MRI studies in standard scan-file (DICOM) format, digitised biopsy slides, structured clinical information, operative and pathology reports, and follow-up records; secondary data comprise public benchmarks and national incidence statistics [9,10]. MRI is stored in an Orthanc server [108], de-identified, and structural scans defaced to reduce re-identification risk [110,111]; slides are de-identified and managed using OpenSlide [117]; clinical data are held in a PostgreSQL database using HL7 FHIR [109] with terminology based on the 2021 classification [36]. Reference annotations are produced by trained annotators using 3D Slicer [94] and QuPath [118], with clinical review and double annotation of a subset to assess agreement; diagnostic labels come from clinician-confirmed records. Automated quality control assesses scan and slide quality, failed records are quarantined, and access is role-based and fully audited. Collection begins only after ethics approval, and in federated experiments only model parameters are shared.

3.9 Data Analysis

Model evaluation uses patient-level stratified cross-validation for development and a held-out local test set, untouched during development, for final reporting. Splits are stratified by tumour type and site so a model is never evaluated on a patient whose other studies were seen in training; where site-level generalisation is the question, leave-one-site-out evaluation is added, since that reveals the scanner and protocol dependence of Section 2.3 [54]. H1 and H2 compare risk-coverage curves and their areas against each baseline, with paired bootstrap intervals from resampling patients [137]; the primary safety comparison, missed-escalation rate at matched coverage, uses McNemar's test [139]. H3, H4 and H5 use paired predictions over the same test cases: the DeLong method for areas under the curve [129], McNemar's test for paired accuracy [139], and the Wilcoxon signed-rank test for per-case metrics such as Dice. H6 uses mixed-effects models with reader and case as crossed random effects. Agreement between AI- and clinician-derived measures is assessed by Bland-Altman analysis, since correlation measures association and not agreement [130]. Family-wise error is controlled per work package by Holm's procedure [138], with endpoints designated in advance. Interviews are analysed by reflexive thematic analysis [134] and triangulated against the escalation data. Analysis uses Python and scikit-learn [107], reported to the applicable standards [57,96].

Each work package's outputs are defined before data collection rather than chosen afterwards. WP2 produces per-case metrics with confidence intervals, reliability diagrams and calibration error for each uncertainty method, an ablation isolating self-supervised pre-training and domain adaptation, and a failure analysis assigning every case in the worst-performing decile to an error category; the taxonomy is itself a reported result, because it identifies where a competence boundary lies. WP3 produces risk-coverage curves for the policy and each baseline, the workload-safety frontier, its sensitivity to the assumed cost of error, and a decomposition of escalations by trigger (low confidence, distributional novelty, conflicting evidence, mandatory review); a second analysis traces error propagation through the agent chain with and without uncertainty propagation, the direct test of H2. WP4 produces concordance for tissue alone, imaging alone and their combination, with an ablation under simulated missing sources. WP5 produces the federated-versus-centralised-versus-single-site comparison and the calibration change after aggregation. WP6 produces geometric measures of planned trajectories with neurosurgeon ratings, deliberately descriptive and formative. WP7 produces the mixed-effects comparison of assisted and unassisted

workflows, the usability and trust scores, and the qualitative themes. Negative results are committed to in advance and reported in the same detail as positive ones [57].

3.10 Ethical, Privacy and Governance Considerations

Ethical approval will be obtained from the relevant Ethics Review Committee of the University of Sri Jayewardenepura and from each participating hospital before any patient data are accessed, and data use will commence only once all approvals are granted. Informed-consent procedures will follow the requirements of the relevant committees and institutions. At ingestion, robust de-identification is applied: removal of direct identifiers, scrubbing of scan-file headers, defacing of structural MRI, and removal of identifying metadata from digitised slides. Privacy is built into the architecture from the outset: where federated learning is used, raw clinical, imaging and tissue data remain within the respective hospitals and only model updates are shared, with role-based access control, authentication and comprehensive audit logging.

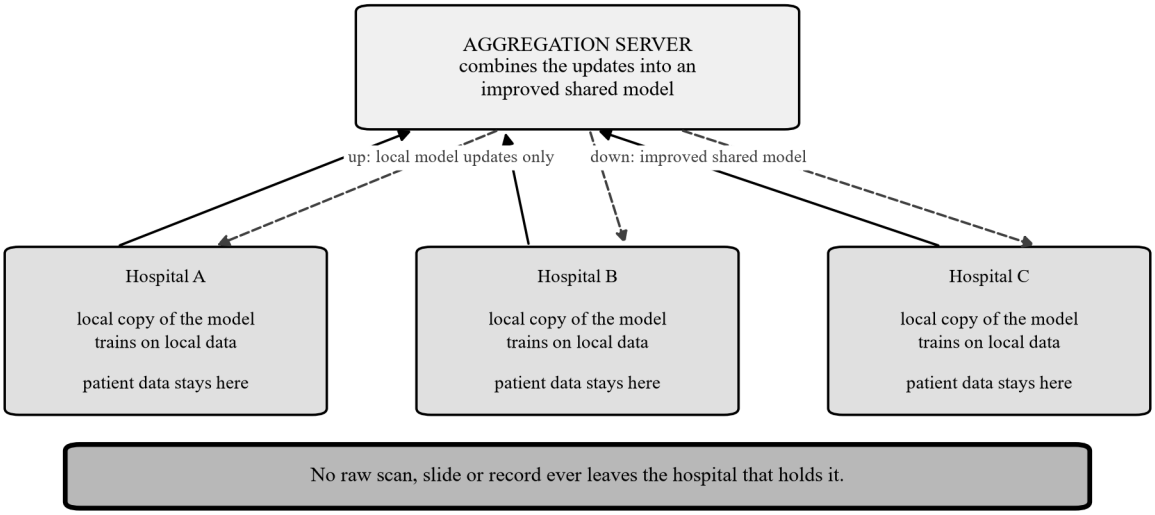


Figure 3.4: Federated training across participating hospitals. Only model parameters are exchanged; raw patient data never leaves an institution.

The system functions strictly as a clinical decision-support tool and will not independently issue or implement unvalidated clinical decisions. All AI outputs require review and validation by appropriately qualified clinicians, and the escalation policy adopts a conservative safety principle whereby cases with insufficient confidence are referred to human experts rather than acted upon automatically. The system supports surgical planning and decision-making only; it will not perform, control or autonomously execute any surgical procedure. All surgical decisions and actions remain under the direct responsibility of the neurosurgeon, who independently reviews and validates the system's outputs before implementation.

3.11 Technology Stack

The stack is chosen for reproducibility, clinical interoperability and mature open-source implementations, so effort goes to the research mechanism rather than infrastructure. Model development uses Python with PyTorch [105] and MONAI [106]; outlining baselines use nnU-Net [43] and Swin UNETR [63]; classical machine learning and metrics use scikit-learn [107]. Image preprocessing uses N4ITK bias-field correction [112], ANTs for registration [113], HD-BET for brain extraction, validated specifically on brain-tumour MRI [114], and FSL and MRtrix3 for nerve-pathway reconstruction [115,116]. Digital pathology uses OpenSlide [117], QuPath [118], stain normalisation following Macenko and Vahadane [119,120], and attention-based multiple-instance learning following CLAM [45,46], with pathology foundation encoders as frozen feature extractors [84,85]. Federated training uses NVIDIA FLARE [121] with Flower and OpenFL as alternatives [122,123], implementing FedAvg, FedProx and SCAFFOLD [49,88,89], and Opacus where differential privacy is required [124,90]. Surgical-path planning uses the Open Motion Planning Library [125,92] with 3D Slicer [94]. The registry is a web application with a PostgreSQL database, an Orthanc imaging server [108], and an HL7 FHIR resource model so it interoperates with national systems rather than becoming another silo [109]. Experiment tracking, dataset versioning and containerisation let every reported figure be regenerated from a recorded model version, data version and seed.

4. Research Plan and Timeline (36 Months)

Table 1: Gantt Chart

Task	Activities	1st Year			2nd Year			3rd Year		
		04	08	12	16	20	24	28	32	36
1.0	**Literature review and proposal refreshment**									
1.1	Literature review and proposal	■	■							
1.2	Ethics application and approval		■	■						
2.0	**WP1: Registry Platform & Data Pipeline**									
2.1	Development of the brain-tumour registry and database and data architecture		■	■	■	■				
2.2	Data ingestion; de-identification;			■	■	■	■			
2.3	Data-quality control; establishment of the research data pipeline.					■	■			
3.0	**Data Collection, Annotation & Quality Control**									
3.1	WP2 MRI perception baselines; calibrated uncertainty; local adaptation.		■	■	■	■	■			
4.0	WP3: uncertainty-aware escalation & orchestration design and experiments.		■	■	■	■	■			
5.0	WP4 computational pathology + multimodal diagnosis & staging.		■	■	■	■	■			
6.0	**Federated Learning Across Hospitals**									
6.1	federated learning across hospitals						■	■	■	■
7.0	Surgical Planning Proof of Concept								■	■
8.0	Thesis writing and publication								■	■

The plan front-loads approvals and the enabling platform, places the core scientific contribution in Year 2, and reserves Year 3 for combined diagnosis, federation, the surgical-planning demonstrator and clinical validation. Third-tier work (federated learning and the surgical-planning demonstrator) is scheduled last so that it can be narrowed without endangering completion.

Table 1: Gantt chart of the 36-month plan.

5. Expected Outcomes and Publications

1. A deployable, doctor-validated, AI-assisted national brain-tumour registry serving the participating units.
2. A validated uncertainty-aware, human-in-the-loop multi-agent framework generalisable beyond this clinical domain.
3. An open, de-identified Sri Lankan brain-tumour imaging and pathology benchmark with an evaluation protocol.
4. At least four peer-reviewed outputs: (1) data-efficient local MRI perception; (2) the core uncertainty-aware escalation and orchestration framework (flagship); (3) computational pathology and combined diagnosis and staging; (4) federated learning across Sri Lankan hospitals. A clinical-validation paper and a surgical-planning proof-of-concept paper are possible further outputs.

6. Evaluation and Success Criteria

Success is judged against measurable criteria rather than a single accuracy figure: (i) perception and pathology models reach clinically useful, well-calibrated performance on held-out local data; (ii) the escalation policy achieves a favourable workload-safety frontier, reducing clinician review load at a controlled and acceptable risk level relative to baselines; (iii) combining imaging and tissue evidence improves diagnosis and staging concordance over single-source baselines; (iv) federated training approaches centralised performance within a quantified margin while sharing no raw data; (v) clinicians rate the assisted workflow, including surgical-planning recommendations, as trustworthy and useful; and (vi) every autonomous action in deployment is traceable and doctor-validated.

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